INCH-POUND

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PERFORMANCE SPECIFICATION

ULTRA HIGH STRENGTH STEEL POWDER FOR ADDITIVE MANUFACTURING FABRICATION OF PARTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for an ultra-high strength steel feedstock material that is used for making parts by additive manufacturing for use on systems and support equipment.

1.2 Classification. Ultra high strength steel powder should be of the following types, as specified (see 6.2).

1.2.1 Type I. For application with powder-Directed Energy Deposition (DED) and Electron Beam melting Powder Bed Fusion (EB-PBF).

1.2.2 Type II. For application with Laser Powder Bed Fusion (LPBF).

Comments, suggestions, or questions on this document should be addressed to Director, Combat Capabilities Development Command (CCDC) Army Research Laboratory (ARL), Weapons and Materials Research Directorate, CCDC ARL Specifications & Standards Office, ATTN: FCDD-RLW-MC, Aberdeen Proving Ground, MD 21005-5069. Since contact information may change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A FSC 9630

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

FEDERAL STANDARDS

FED-STD-313	-	Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

(Copies of this document are available online at https://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A751	-	Standard Test Methods, Practices, and Terminology for Chemical Analysis of Steel Products
ASTM B212	-	Standard Test Method for Apparent Density of Free-Flowing Metal Powders Using the Hall Flowmeter Funnel
ASTM B213	-	Standard Test Methods for Flow Rate of Metal Powders Using the Hall Flowmeter Funnel
ASTM B214	-	Standard Test Method for Sieve Analysis of Granular Metal Powers
ASTM B215	-	Standard Practices for Sampling Metal Powders
ASTM B243	-	Terminology of Powder Metallurgy

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ASTM B330	-	Standard Test Methods for Estimating Average Particle Size of Metal Powders and Related Compounds Using Air Permeability
ASTM B822	-	Standard Test Method for Particle Size Distribution of Metal Powders and Related Compounds by Light Scattering
ASTM B859	-	Standard Practice for De-Agglomeration of Refractory Metal Powders and Their Compounds Prior to Particle Size Analysis
ASTM B964	-	Standard Test Methods for Flow Rate of Metal Powders Using the Carney Funnel
ASTM E352	-	Standard Test Methods for Chemical Analysis of Tool Steels and Other Similar Medium- and High-Alloy Steels
ASTM E354	-	Standard Test Methods for Chemical Analysis of High-Temperature, Electrical, Magnetic, and Other Similar Iron, Nickel, and Cobalt Alloys
ASTM E1019	-	Standard Test Methods for Determination of Carbon, Sulfur, Nitrogen, and Oxygen in Steel, Iron, Nickel, and Cobalt Alloys by Various Combustion and Inert Gas Fusion Techniques
ASTM E1131	-	Standard Test Method for Compositional Analysis by Thermogravimetry
ASTM E2651	-	Standard Guide for Powder Particle Size Analysis
ASTM F660	-	Standard Practice for Comparing Particle Size in the Use of Alternative Types of Particle Counters
ASTM F3049	-	Standard Guide for Characterizing Properties of Metal Powders Used for Additive Manufacturing Processes

(Copies of these documents are available from https://www.astm.org)

Society of Automotive Engineers International Standard

SAE Aerospace Material Specification

AMS-2269 Chemical Check Analysis Limits, Nickel, Nickel Alloys, and Cobalt Alloys

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(Copies of these documents are available from https://www.sae.org)

Occupational Safety & Health Administration Regulations - Hazardous Communication Standard - 29 CFR 1910.1200

(Copies of these documents are available from www.OSHA.gov.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS: Powders shall meet the requirements cited in 3.1 thru 3.10 of this specification.

3.1 Chemistry. For types I and II powders, the chemical composition shall conform to the weight percentages presented in table I.

Table I. Chemical composition in weight % of ultra-high strength steel powder feedstock.

Element	Min	Max	Aim
Al	-	0.015	-
C	0.260	0.290	0.270
Cr	2.50	2.80	2.65
Cu	-	0.20	-
H	-	0.0002	-
Mn	0.50	0.80	0.65
Mo	0.85	1.05	0.95
N	-	0.0070	-
Ni	0.90	1.20	1.00
O	-	0.0100	-
P	-	0.01	-
S	-	0.005	-
Si	0.90	1.10	1.00
Ti	-	0.006	-
V	0.05	0.15	0.10
P+Sn+As+Sb	-	0.035	-
Fe		balance	

Chemical composition and characterization shall be determined by property type techniques allowed in ASTM F3049, E354, E1019, spectrographic methods, or by other applicable methods approved by CCDC Army Research Laboratory (ARL) or applicable government Cognizant Engineering Authorities (CEAs).

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3.2 Chemical composition. The chemical composition of the powders shall be the same as that of the sample furnished for first article testing within the tolerances allowed in this specification. Any change in the formulation of a qualified powder that affects the powders morphology, chemical composition, material condition, and/or contaminant level shall require re-evaluation. Types I and II shall have powder samples to validate chemistry as laid out in 3.1 by elemental analysis. Special care shall be made to ensure that the oxygen, nitrogen, and hydrogen content shall not exceed that set in table I.

3.2.1 New powder. When purchasing new powder, the powder shall be tested to meet the requirements of elemental composition set in 3.1 and powder properties set in 3.4.

3.2.2 Powder reuse. Recyclable material may be utilized with condition that it still meets all of the requirements in this specification.

Type I: After manufacturing, powder shall be sieved with a 100 mesh screen. Powders can be reused if they meet the requirements for the chemical composition, particle size distribution, and morphology as set in 3.1 and 3.4.

Type II: After manufacturing, powder shall be sieved with a 200 mesh screen. Powders can be reused if they meet the requirements for the chemical composition, particle size distribution, and morphology as set in 3.1 and 3.4.

3.3 Manufacturing method. Manufacturing shall occur via gas atomization under an inert argon gas environment using gas of purity sufficient to meet impurity requirements specified in table I. The manufacturing processes for the production of the powders shall be as specified in the contract or purchase order and shall produce a powder that meets or exceeds the requirements of 3.1 thru 3.9.

3.4 Properties. The material shall meet the following properties requirements before use.

3.4.1 Particle shape. The particle morphology shall be spherical determined in accordance with 4.7.6 through optical inspection via such methods such as SEM or calculated through optical imaging. Powder shall be free of contamination.

3.4.2 Particle size distribution. The particle size distribution for all powders shall be made available to the applicable Cognizant Engineering Authority (CEA) or Engineering Support Activity (ESA) as measured in accordance with 4.7.3 or as specified in the contract or purchase order. The particle size distribution shall fall within 10% of the originally required distribution for D10, D50, and D90 values in order to fulfill conformance requirements.

3.4.3 Oxygen content. The oxygen content shall meet the requirement as shown in table I.

3.4.4 Flowability. The powder shall be flowable by using any method that is workable and the results are only for guidance.

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3.4.5 Apparent density. Powder apparent density, Hall (Apparent Density Hall) shall be 2.5 g/cm3 minimum, determined in accordance with ASTM B212.

3.4.6 Quality / Agglomerates. All powders, as received by the purchaser, shall be thoroughly blended, dry, and free from foreign materials, clumps (agglomerates) and individual particles no larger than 150 microns for type I and 50 microns for type II as determined in accordance with 3.4.2 and 4.7.4. Powder shall be free of contamination.

3.5 Storage and Handling. Material shall be stored in sealed, airtight container under an inert environment. The containers shall be capable of protecting the powder against damage or contamination until it is ready for use. Material shall be handled while wearing appropriate Personal Protection Equipment (PPE) and as required from the Safety Data Sheet.

3.6 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.4 (see 6.2).

3.7. Material. The powders that are supplied under this specification shall be as specified herein. Powders not specified shall be selected by the contractor and shall be subjected to all the provisions of this specification and meet the requirements cited in section 3 of this specification. An evaluated product shall be considered acceptable when a certified test report(s) showing that the material conforms to all the requirements of this specification (3.1 to 3.9).

3.8 User instruction markings. All primary containers shall be legibly marked or labeled, noting the material, particle size, lot number of the alloy powder, and any special markings specified by the contract or purchase order (see 6.2).

3.9 SDS. The manufacturer shall comply with the requirements as specified in the OSHA Brief (see: https://www.osha.gov/dsg/hazcom/) and as specified in Appendix D of 29 CFR 1910.1200).
(see: https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.1200AppD).
A SDS shall be prepared for the material in accordance with FED-STD-313 and forwarded to the end user and contracting activity (see 6.2). The SDS shall be included with each shipment of the material covered by this specification as required (see 6.2) and submitted to pertinent Government agencies as stated in FEDSTD-313.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

First article inspection and tests (see 4.4).

4.2 Testing responsibility and facilities. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all the requirement tests as specified herein. Unless otherwise specified in the contract or purchase order, the contractor may use his

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own or any other facilities suitable for the performance of the requirement tests specified herein. The Government reserves the right to perform or check any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements and to determine the validity of the certifications.

4.3 Inspection lot. A lot of material shall be defined as the amount produced of the same composition or composition blend, the same particle size, and the same condition and manufactured at the same time from the same batches of basic materials by the same manufacturing process, and submitted for vendor's inspection at the same time. The addition of any substance including composition blend to a batch shall constitute a new lot.

4.4 First article inspection. The first article samples submitted in accordance with section 3 shall be examined for all the provisions of this specification applicable to end item examination and shall utilize the same requirements and test methods as the production acceptance inspection. Unless otherwise specified (see 6.2), the supplier shall prepare test powder for testing of 4.7.1 to 4.7.6.

4.4.1 First article tests. First article tests shall consist of all the tests specified in 4.4.

4.5 Noncompliance. If a sample fails to pass its first article inspection, the lot shall be rejected. Further guidance for noncompliant samples shall be specified in the contract or purchase order (see 6.3).

4.6 Sampling. Test samples shall be taken from the master lot in accordance with ASTM B215, Practice B. Sufficient powder shall be taken from each inspection unit to perform at least three (3) evaluations for each test.

4.7 Test methods.

4.7.1 Chemical composition. Samples for chemical analysis (see 3.1) shall be prepared and tested in accordance with ASTM A751 and E352, SAE AMS-STD-2269 or by one of the following methods; Inductively Coupled Plasma - Optical Emission Spectrometry (ICP-OES) or Inductively Coupled Plasma - Mass Spectrometry (ICP-MS) or as specified in the contract or purchase order.

4.7.2 Non-metallic impurities.

4.7.2.1 Moisture and volatiles. The wt. % of the non-metallic impurities, moisture and volatiles shall be measured by the thermogravimetric method in accordance with ASTM E1131. For this test only inert gas purge shall be used, and the maximum temperature shall be 500° C. Maximum allowable non-metallic impurities and moisture are covered in table I.

4.7.2.2 Oxygen content. Oxygen content (see 3.1 and table I) shall be determined by inert gas fusion, as described in ASTM E1019 or as specified in the contract or purchase order.

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4.7.3 Particle size distribution. Powder (see 3.4.2) shall be tested in accordance with ASTM B214 (sieve analysis) or ASTM B822 (light scattered particle size distribution) or optical (ASTM E2651 and ASTM F660 or any approved CCDC ARL testing) or as specified in the contract or purchase order.

4.7.4 Agglomerates / Quality. The size of the agglomerates shall be measured in accordance with practice guidelines found in the ASTM standards B859 and B330. Optical or electronic microscopes may be used. The dry powder sample preparation technique shall be used. As an alternative to the microscopy-based technique, a sieve analysis can be used. If an U.S. standard sieve analysis is used, then a 99.5% of a 100 gram sample of the powder shall pass through a 100 mesh for type I and a 200 mesh for type II.

4.7.5 Flowability. Any test method workable shall be used, such as a Hall flow meter (ASTM B213) or a Carney flow meter (ASTM B964) with the understanding that specific characteristics of the machine and in certain situations will produce mixed results and only for guidance.

4.7.6 Particle shape. Particle shape shall be determined through optical inspection via such methods such as SEM or calculated through optical imaging by following ASTM B243 (see 3.4.1).

4.7.7 Quality. The requirements specified in section 3.4.5 shall be determined by visual inspection or as specified in the contract or purchase order. For chemical analysis, the manufacturer shall provide and maintain a certified quality assurance system required by this specification, covering the supplies under the contract or purchase order.

4.8 SDS. The SDS shall address all components of the powder and be in compliance with FED-STD-313. Noncompliance to 3.10 shall be cause for rejection.

4.9 Manufacturer's certification. The manufacturer shall certify and maintain substantiating evidence (Objective Quality Evidence) that the material meets the requirements of this specification and that the powder conforms to the manufacturer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery or provided for under the provisions of the contract.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order. When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

Information for Guidance Only. (This section contains information of a general or explanatory nature that is helpful, but is not mandatory.)

6.1 Intended use. The material covered by this specification is intended for use in the manufacture of ultra-high strength steel powder for additive manufacturing of parts.
6.2. Ordering data. Acquisition documents should specify the following

a. Title, number and date of this specification
b. Specify type of powdered alloy required (see 1.2)
c. If issues of documents are different (see 2.2.1 and 2.3)
d. If first article inspection is different (see 3.6 and 4.4)
e. Selection of applicable level of markings and packaging requirements (see 3.8 and 5.1)
f. An SDS is required with each shipment (see 3.9)
g. Unit container and size if other than specified (see 5.1)

6.3 Non-Compliance. If a sample fails to pass its conformance inspections, the lot should be rejected and the manufacturer should notify the cognizant inspection or contracting activity of such failure and take corrective action on the materials or processes, or both, as warranted. Acceptance and shipment of the product should be discontinued until corrective action, suitable to the inspection activity, has been taken. After the corrective action has been taken, conformance inspections should be repeated on the new lot. In the event of failure after re-inspection, information concerning the failure should be furnished to the cognizant inspection or contracting activity.

6.4 SDS. Contracting officers will identify those activities requiring copies of a completed SDS prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.5 Objective Quality Evidence. A documented statement of fact, or supporting record, either quantitative or qualitative, pertaining to the quality of an item or process, based on observations, measurements, or tests that can be verified as required within the document. Evidence will be expressed in terms of specific quality requirements or characteristics in a certified lab report. These characteristics are identified in drawings, specifications and other documents which describe the item, process or procedure.

6.6 Certification. A certified test report from an ISO 9000/IEC 17025 or NADCAP certified laboratory should show that the powder conforms to all the requirements of the specification. Be prepared to offer evidence of the certification if requested.

a. National Aerospace and Defense Contractors Accreditation Program (NADCAP) Performance Review Institute (PRI) accreditation

(Society of Automotive Engineers International / Aerospace Quality System AC7004) program requirements

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Performance Review Institute (PRI) PD 1100

(Copies of these documents are available from https://p-r-i.org.)

b. International Organization for Standardization (ISO) Publications American National Standards Institute

ISO 9000 - Quality Management Systems - Fundamentals and Vocabulary. Standards

ISO/IEC17025 - General Requirements for the Competence of Testing and Calibration Directives Laboratories.

(Copies of these documents are available from https://www.iso.org/home.html.)

ISO 9001/2015 or ISO/IEC 17025 Accreditation. See: https://www.iso.org/home.html.

In reviewing and validating certification by the Engineering Support Activity (ESA), laboratories certified should submit their accreditation from the external certification body or registrar.

6.7 Cognizant Engineering Authority. The CEA assumes all responsibility for assuring that all required testing and subsequent evaluation of any powders are completed and pass all requirements specified herein, and also meet any additional performance criteria specific to their intended application.

6.8 Engineering Support Activity. The ESA provides technical data, quantities, packaging requirements, possible suppliers, special handling, storage requirements, and shelf life information to procure the powder and resolve individual engineering support problems.

6.9 Subject term (key word) listing.

Composition

Flowability

Impurities

Laser

Material

Particle

Quality

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CONCLUDING MATERIAL

Custodians:
Army - MR
Navy - AS

Preparing Activity:
Army - MR

(Project 9630-2020-001)

Review activities:
Army - AV, CR
Navy - AS, OS, SH
DLA - CQ, DH

Civil agency:
GSA - FAS

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